QQ-L-00171b(NAVY SHIPS)
15 August 1956

INTERIM FEDERAL SPECIFICATION

LEAD PIG

This interim Federal specification has been developed by the Department of the Navy, Bureau of Ships, based upon currently available technical information. It has not been approved for promulgation as a revision of Federal Specification QQ-L-171a, dated 18 June 1952, and is subject to modification. This specification is in interim form only, and pending its promulgation by the Administrator, General Services, as a regularly coordinated Federal specification, it may be used in procurement.

1. SCOPE

1.1 Scope. - This specification covers lead pigs of the composition specified in 3.3 (see 6.1).

1.2 Classification. - The pigs shall be furnished in commercial standard forms and shapes, in the following grades as specified (see 3.3 and 6.2):

Grade A
Grade B
Grade C

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
NN-P-71 - Pallets; Meterials-Handling, Wood (General Construction Requirements).
QQ-M-151 - Metals; General Specification for Inspection of.
QQ-S-781 - Strapping, Flat; Steel.

FSC 9650

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- The lead pigs shall be made from ore or other material by processes of reduction and refining. For grades A and B, reclaimed lead, obtained by recovery of metallic lead and its alloys by the simple reclaiming process of melting, drossing, and casting is acceptable under this specification if chemical composition requirements are within limits specified herein. Grade C lead pig shall be produced from ore or other materials by a conventional process of reduction and refining or other processes which equally preclude the presence of unusual deleterious elements. Reclaimed lead is not acceptable for grade C.

3.2 Surface condition.- The pigs shall be clear and reasonably free from adhering foreign matter.

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3.3 Chemical composition.-

3.3.1 Lead pigs shall conform to the chemical requirements shown in table I.

Table I - Chemical requirements.

Grade	Silver (max.)	Copper (max.)	Arsenic (max.)	Antimony (max.)	Tin (max.)	Zinc (max.)	Iron (max.)	Bismuth (max.)	Lead (min.) [1]
	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent
A	[2]	[2]	[2]	[2]	[2]	[2]	[2]	[2]	99.90
B	[3]	[3]	[3]	[3]	[3]	[3]	[3]	[3]	95.00
C	0.02	0.040 to 0.080 [4]	5	5	5	0.001	0.002	0.025	99.90

[1] By difference.
[2] Total impurities 0.10 percent maximum.
[3] Total impurities 5.000 percent maximum.
[4] Range.
[5] Arsenic, antimony, and tin together - 0.002 percent maximum.

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3.3.2 The contractor shall furnish an analysis of each melt showing the percentage of each of the elements specified in 3.3.1.

3.4 Identification marking.- Each pig shall have the following information cast or stamped thereon in legible characters: The Federal Specification symbol, grade, the manufacturer's name or brand, and melt or cast number.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Sampling, inspection, and tests shall be conducted in accordance with Specification QQ-M-151 and as specified herein.

4.2 Sampling.-

4.2.1 Lot.- A carload or fraction thereof of pig lead of the same grade shall constitute a lot for sampling.

4.2.2 Sampling for visual examination.- A random sample of lead pig shall be selected from each lot offered for delivery and examined for visual characteristics in accordance with Standard MIL-STD-105 at inspection level I. The Acceptable Quality Level shall be 2.5 percent defective (see 4.4).

4.2.3 Chemical analysis.- One pig shall be taken by the inspector to represent each ton of lead in the lot and the pig or pigs thus taken sampled by any one of the methods specified hereinafter. Except when the spectrographic analysis is to be made on a solid sample (see 4.2.3.5), not less than 600 grams from each lot shall be forwarded for analysis to a laboratory designated by the bureau or agency concerned (see 4.3.2).

4.2.3.1 Sawing.- The pigs shall be sawed completely through in the center, at each end of largest cross section, and at points midway between ends and center. The sawings shall be mixed thoroughly, quartered, and samples for analysis taken from the mixed materials. Sawings shall be free from all extraneous material and shall be treated with a strong magnet in order to remove iron introduced from the saw blade.

4.2.3.2 Drilling.- The pigs shall be drilled preferably with a 1/2 inch drill at least halfway through from two opposite sides. In drilling, the holes are spaced along a diagonal line from one corner of the pig to the other. If several pigs are used, the holes may be drilled in each of the several pigs of a group so that holes represent consecutive positions on the diagonal of a single pig. The drillings may

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be clipped and mixed and samples for analysis drawn from mixed material; or they may be melted in a clean crucible and either granulated by pouring into distilled water and drying thoroughly, or cast into thin slabs which shall be sawed through completely in several places and sawings treated as specified in 4.2.3.1.

4.2.3.3 Kettle.- Unless otherwise agreed upon by the bureau or agency concerned and the contractor, the pigs or portions of lot selected shall be melted in a clean kettle. The melt shall be made at as low a temperature as possible, preferably not over 685°F., in order to prevent excessive drossing, and the molten bath shall be stirred immediately prior to sampling. While stirring, a sample in a preheated ladle may be taken and cast into thin sample bars approximately 8 inches in length, 1-1/4 inches in width and 3/8 inch in thickness. From the bar so cast, saw cuts shall be made 1/2 inch apart alternately from each side, halfway across the slab. No cut shall be made closer than 1 inch to either end of the slab. The sample shall be treated with a strong magnet to remove iron.

4.2.3.4 When mutually agreed upon between the bureau or agency concerned and the contractor, ladle test ingots obtained during pouring of the metal shall be used as samples and treated as specified in 4.2.3.1, 4.2.3.2, and 4.2.3.3. One ladle test ingot shall be taken at the beginning of the pouring of a lot and one near the end of pouring as indicated by the inspector.

4.2.3.5 Spectrographic analysis.- Samples for spectrographic analysis shall be obtained by pouring metal representative of the heat or melt into a permanent mold in the presence of the inspector or at the time castings are poured. The mold shall be so designed that the samples obtained will be truly representative of the heat or melt and satisfactory for use by the laboratory at which the spectrographic analysis is to be performed.

4.3 Inspection and tests.-

4.3.1 Visual examination.- Each of the sample lead pigs selected in accordance with 4.2.2 shall be visually examined to verify compliance with the requirements of this specification regarding surface condition and identification marking (see 3.2 and 3.4).

4.4 Chemical analysis.- Chemical analysis shall be made on each sample selected in accordance with 4.2.3 by wet chemical or spectrographic methods to determine conformance with 3.3. In case of dispute, chemical analysis by wet chemical methods shall be basis for acceptance.

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4.5 Rejected lots.- Any lead pig in the sample containing one or more defects shall be rejected, and if the number of defective lead pigs in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming lead pigs. Samples shall again be selected from such resubmitted lots and examined to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.2).-

5.1.1 Level A and level B.- Lead pig shall be packed on pallets conforming to type III of Specification NN-P-71. The stringer ends shall be bevelled. The load shall be adequately tension tied to the pallet with 1-1/4-inch by 0.035-inch coated metal strapping conforming to Specification QQ-S-781. Full length, nominal 1 by 4 inch lumber shall be placed under the strapping at the top of the load. The gross weight shall not exceed 2,240 pounds.

5.1.2 Level C.- Lead pigs may be shipped in bulk. In all cases pigs of each grade shall be segregated. If special packing is necessary to insure the acceptance of lead pigs by common or other carriers for safe transportation at the lowest rate to the point of delivery, such packing shall be provided by the contractor. Pigs when presented for inspection, or loaded for shipment, shall be properly separated by grade.

5.2 Marking.-

5.2.1 Military.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

5.2.2 Non-Military.- Shipping containers shall be marked with the name of the material, the grade of pigs contained therein as defined by the contract or order under which shipment is made, the name of the contractor, the number of the contract or order, the specification symbol, and the Government stock numbers.

6. NOTES

6.1 Intended use.

6.1.1 Grade A.- Grade A lead pig is intended for foundry and chemical uses.

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6.1.2 Grade B.- Grade B lead pig is intended for uses such as weights and ballast.

6.1.3 Grade C.- Grade C lead pig is intended for use in radiation shielding.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Grade of lead pig required (see 1.2).
(c) Whether level A and B or level C packing is required (see 5.1).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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